

UNIVERSITY COLLEGE LONDON

DEPARTMENT OF PHYSICS & ASTRONOMY

High Resolution Rovibronic Spectroscopy of I_2

RESEARCH ESSAY

Author: Caleb Burhan

Supervisor: Jonathan Tennyson

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1 Introduction

This literature review will cover the spectroscopy of $^{127}I_2$. I will start with a discussion of the use of iodine for electric propulsion and the importance that spectroscopic information has on modelling iodine plasmas for this purpose. I will then provide an outline of the important use of the spectrum of molecular iodine as a frequency standard for laser calibration and as a reference for spectroscopic experiments. Section 2 will cover the molecular structure of I_2, including the prediction and first identifications of the main electronic states and spectroscopic systems. Section 3 will discuss the important spectroscopic investigations in the literature, with a focus on papers which provide an assigned list of transitions. The future scope discussion will argue that the wealth of available high resolution spectra investigations for the purpose of wavelength standards justifies performing an inversion process to produce energy levels with associated uncertainties from the measured transitions. The process to be undertaken using some of the data covered in this review is the *measured active rotational-vibrational energy levels* (MARVEL) system of Furtenbacher *et al.* [1].

1.1 Iodine as a propellant for Electric Propulsion

Electric propulsion (EP) is a system of spacecraft propulsion in which a propellant is accelerated using electrothermal, electrostatic, or electromagnetic methods in order to produce thrust[1] [2]. The energy source and the propellant, which is the mass ejected from the engine in order to create an impulse, are uncoupled in an electric propulsion system. In a traditional chemical propulsion system however, the propellant is accelerated by its own chemical energy. As a result of this, electric propulsion achieves higher propellant exhaust speeds and in turn lower rates of propellant consumption (with the drawback of low thrust levels [2]). This makes electric propulsion systems ideal for in-space use, such as satellite orbit corrections and space exploration probes.

Xenon is by far the most commonly used propellant in EP systems. However xenon is rare and expensive to produce commercially [3]. Iodine is a viable alternative due to the low ionisation potential and high mass of both I and I_2 [4], both important properties which determine the thrust-to-power ratio [5]. Iodine is a solid at room temperature and is therefore easier to store than a gas such as xenon, and an amount of iodine needed for propellant usage can be procured for less than 1/20 of the cost of the same amount of xenon [3]. Table 1 is a summary of different propellants with data taken from a topical review of electric propulsion [2]. Note that iodine is the only option given with a molecular component, as energy is lost in exciting molecules into internal vibration and rotation

[1] Plasma thrusters and ion-propulsion are used to refer to the same systems.

Table 1: Comparison of EP propellants taken from a topical review [2].

Propellant	Mass (amu)	E_i (eV) [a]	State	Cost
He	4.0	24.6	Gaseous	Low
Li	6.9	5.4	Solid	Low
Ar	39.9	15.8	Gaseous	Low
Kr	83.9	14.0	Gaseous	High
Xe	131.3	12.1	Gaseous	Very high
Cs	132.9 [b]	3.9	Liquid	Very high
Hg	200.6	10.4	Liquid	Low
Bi	209.0	7.3	Solid	Low
I_2 (I)	253.8 (126.9)	9.4 (4.1)	Solid	Low

[a] The first ionisation energy
[b] A surely mistyped or misprinted value of 1.2.9 is given in the original table. Above value taken from the NIST WebBook [6].

modes. The benefits of iodine outweigh this general drawback of molecules [2].

The first in-space demonstration of an iodine propulsion system was performed in 2021, with manoeuvres confirmed with satellite tracking data [3]. It was demonstrated that the use of pressurised propellant enabled a smaller system design. A propulsion system miniaturised enough to be used by the growing number of small satellites would ensure that these satellites are able to avoid collisions and de-orbit safely at end of life. This is important in mitigating the long term issue of space debris build up [3].

Improvements for iodine propulsion systems require extensive modelling of the behaviour of iodine plasmas, which in turn require, amongst other things, extensive and accurate data on reaction cross sections and absorption spectra for atomic and molecular iodine [5, 7–9]. In [5], absorption spectra were used to track the densities of neutral atomic and molecular iodine in a plasma over various temperatures and pressures. It is partly the molecular nature of iodine which increases its ionisation costs at high pressures which effects its performance as a propellant [8]. It is these studies which motivate further analysis of existing spectroscopic studies of iodine, and the production of accurate vibration-rotation energy levels.

1.2 Iodine as a frequency standard

The spectrum of molecular iodine has often been used as a frequency standard in the visible and near ultraviolet. Many of the high resolution spectroscopic investigations of molecular iodine have been for the purpose of providing reference wavelengths for laser frequency calibration and stabilisation, both over broad “atlases” covering a large bandwidth [10–14] and hyperfine resolved analyses of specific rotational lines [15–17].

Tellinghuisen cites a 1976 paper [18] as an early example of the I_2 B-X spectrum being used as a wavelength calibrator [19]. The study in question however, was investigating an HeI_2 van der Waals complex. The complex was formed in a jet of helium and I_2 gas, and the florescence band features of the HeI_2 complex were compared to the associated I_2 bands to discern the amount of I_2 which was forming a mixture with the helium gas. It was not so much an example of the spectrum of I_2 as a wavelength calibrator but an investigation involving the molecule directly.

The following are examples of spectroscopic investigations of other systems which have used the spectrum of iodine taken from an atlas as a standard from which to calibrate equipment or adjust results. A study of benzene–Argon van der Waals clusters [20] used simultaneous measurement with iodine to aid the vibrational analysis of the measured spectra. In a Fourier-Transform spectroscopic investigation of water [21], producing data later used in a MARVEL study [22], an iodine cell was placed in the light path for one of the recorded spectra in order to compare line positions. Comparison with the Luc and Gerstenkorn atlas [10, 23] led to the recorded spectra being shifted and then scaled. In a recent spectroscopic study of YbOH as part of a search for beyond standard model physics, the iodine atlas of Salami and Ross [24] was used to calibrate data by sending a portion of their cw laser through an iodine cell.

The use of specific high-resolution iodine transitions for laser frequency stabilisation is driven by the proximity of certain known lines with hyperfine resolved measurements to commonly used lasers. In 1969, Hanes and Dahlstrom [25] discovered a coincidence of hyperfine components of the R(127) line of the (11, 5) band of the I_2 B-X transition with the radiation of a 633 nm helium-neon (He-Ne) laser. It was proposed that these components could be used for laser frequency stabilisation. Around the 532 nm wavelength region, a frequency-doubled Nd:YAG can be tuned to multiple hyperfine resolved iodine lines to allow for calibration [17, 26, 27]. The hyperfine resolved transitions around the 532 nm wavelength region from [17] were used as a frequency network to demonstrate an optical clock. The frequency stability of the optical clock was transferred via a frequency comb to a broader bandwidth domain. The authors were able to demonstrate a portable iodine cell based clock with a stability rivalling only massive national-standards based installations, with a relative measurement uncertainty of 4×10^{-13} over an entire year when compared to a Cs clock standard [28]. The portability of iodine based laser stabilisation equipment has lead to its implementation for laser stabilisation in space [29], also using transitions around 532 nm. Iodine spectra components at both 532 nm and 633 nm, used with the aforementioned lasers, form part of the recommended radiations for frequency stabilisation for the practical realisation of the metre [30, 31].

2 Characterising the Spectra of Iodine

As iodine is a homonuclear diatomic it has no permanent dipole moment, so rotation-vibration structure in the spectra of iodine can discerned only through electronic transitions or with Raman spectroscopy. The $B^3\Pi^+_{0u}$ - $X^1\Sigma^+_{0g}$ dominates the visible spectrum and so these states are the most studied [32], but other states are also studied both through spectroscopy and theoretical calculations.

To start with I will discuss the choice of term symbols and coupling schemes for the valence states of iodine. I will discuss notation for labelling the transitions between rotational and vibrational states, and the effect of nuclear spin statistics on the intensities of rotational lines. I will then look at the first predictions and observations of the main electronic states of iodine, by 'main' I mean states which form parts of investigated and rotationally resolved spectroscopic systems.

2.1 State symbols and good quantum numbers

Vector quantities are denoted by bold type, and I will use units such that $\hbar = 1$. The valence states of iodine fall into the Hund coupling cases (a) and (c) [32]. Hund's case (a) is the non-relativistic case and is used when the orbital angular momentum $\boldsymbol{L}$ is coupled to the internuclear axis, so the orbital angular momentum parallel to this axis, Λ, is a good quantum number [33]. Without Ω for now, the Hund case (a) term symbol is ${}^{2S+1}\Lambda^{+/-}_{g/u}$. The +/- term denotes whether the state wavefunction is symmetric or anti-symmetric with respect to a reflection across any plane containing the internuclear axis [34], this only arises for $\Lambda = 0(\Sigma)/\Omega = 0$ states. g (gerade) and u (ungerade) denote whether the state is symmetric or antisymmetric with respect to a parity transformation of the electronic coordinates about the centre of (charge) symmetry. This arises for any molecule whereby the field created by the nuclei has a centre of charge symmetry [35], which for I_2 as a homonuclear diatomic exists and is the midpoint of the two nuclei. The parity transformation in question is the inversion operator $\hat{\imath}$, which inverts the coordinates of the electrons (but not the nuclei or spins),

$$\hat{\imath}\psi_{el}(x_i, y_i, z_i) = \psi_{el}(-x_i, -y_i, -z_i) = \pm\psi_{el}(x_i, y_i, z_i)$$

where the electronic states corresponding to eigenvalue $+1$ have g parity and those with eigenvalue -1 have u parity [36]. As an example the iodine ground state is designated with the letter X, has total electron spin $S = 0$, and an orbital angular momentum parallel to the internuclear axis $\Lambda = 0$ which is denoted by Σ (in keeping with the convention in atomic physics of *spdf* orbitals). So the ground state is written $X^1\Sigma^+_g$ in this coupling

scheme. The total angular momentum (exclusive of nuclear spin), $\boldsymbol{J}$, has component R perpendicular to the internuclear axis where R is the magnitude of the angular momentum of the rotating molecular frame $\boldsymbol{R}$ (this vector is already perpendicular to the internuclear axis). The component of $\boldsymbol{J}$ parallel to the internuclear axis is Ω, which is the sum of the axial components of $\boldsymbol{L}$ and $\boldsymbol{S}$ ($\Omega = \Lambda + \Sigma$ where Σ in this context is the component of S along the internuclear axis).

It was in fact iodine in particular for which Mulliken [37, 38] noted that a coupling scheme analogous to L-S coupling for atoms was insufficient. A scheme analogous to j-j coupling in atoms, is Hund's case (c) [39]. Hund's case (c) is used when the spin-orbit is strong enough to couple $\boldsymbol{S}$ and $\boldsymbol{L}$ which sum to a resultant $\boldsymbol{J_a}$ (Atkins uses $\boldsymbol{E}$ [33]), this has component Ω along the internuclear axis. This is known as the Ω representation as Ω is used in the main term symbol as opposed to Λ. Ω is still well defined in case (c) despite Λ and Σ not being well defined. The total angular momentum $\boldsymbol{J}$ is obtained by taking $\boldsymbol{\Omega}$ to be a vector of magnitude Ω (the axial component of $\boldsymbol{J_a}$) lying along the internuclear axis; $\boldsymbol{J}$ is then the vector sum of $\boldsymbol{\Omega}$ and $\boldsymbol{R}$. Veseth in 1973 [39] provided a theory paper deriving the rotational energies of a Hund case (c) molecule, along with a way of reducing the case (c) rotational energy expression to that of case (a).

As shown in Figure 2, the A', A, and B states of iodine all have the same term symbol in Hund's case (a), $^3\Pi_u$ (A and A' do not have a +/- symmetry term as this only arises for $\Omega = 0$ states), the Ω representation differentiates symmetry of the states. In the literature they are treated as independent states with their own letter labelling (see section 2.3) instead of just fine structure splittings of the $^3\Pi_u$ state. The ground state in the (c) case is written $X^1 0^+_g$.

When both coupling terms are available I will designate the electronic states following the notation used by Mulliken [38, 40]. An arbitrary term is $x\ ^{2S+1}\Lambda^{+/-}_{\Omega g/u}$ with x a letter corresponding to energy uniquely identifying each state. The standard practice of using uppercase letters for states with the same spin multiplicity as the ground state and lowercase letters for states with different spin multiplicity [34] is not used for iodine. The standard state labels in the literature use uppercase letters with primes and double primes, as well as lowercase letters and Greek letters.

The vibrational bands can be labelled as (v', v''), with v'' the vibrational quantum number for the lower state and v' the vibrational quantum number for the upper state. The notation for a simultaneous change in rotational state uses a letter to designate the branch, meaning the change in rotational quantum number J. The R branch corresponds to $\Delta J = +1$ and the P branch corresponds to $\Delta J = -1$. So the R(127) line of a band corresponds from a transition from a $J = 127$ level to a $J = 128$ level. When $\Delta J = 0$ transitions are allowed they correspond to the Q branch. $\Delta J = \pm 2$ correspond to the S

and O branches, which appear for two photon [41] and Raman transitions [36].

2.2 Nuclear spin statistics

Tolansky [42, 43] confirmed experimentally the nuclear spin of iodine to be $I = 5/2$. Nuclear spin affects the population of rotational state populations of molecular iodine, and thus the rotational structure of the spectra of molecular iodine due to the Pauli principle. The nuclei are Fermions due to their half integer spin, and every half rotation interchanges the identical nuclei. As a result the total wavefunction of the molecule must be antisymmetric with respect to the operator $\hat{P}_{12}$ exchanging the two nuclei. The nuclear component of the total wavefunction can on its own be symmetric or antisymmetric. The only way then to give the correct symmetry for the total wavefunction is by ensuring the rotational state makes the whole wavefunction antisymmetric [2]. So the proportion between the odd J and even J rotational state populations is determined by the statistics of the nuclear symmetry. Using a general rule for a Σ_g^+ molecule derived in [33] for iodine with $I = 5/2$, the proportion of symmetric and anti-symmetric nuclear states is:

$$\frac{N_{symmetric}}{N_{antisymmetric}} = \frac{I+1}{I} = \frac{7}{5}$$

So there should be a 7:5 ratio in population between rotational states with odd and even values of J. The resulting variation in (relative) intensity between the odd and even J-values is reported by Simmons and Hougen [44] and is clearly visible in the spectrum provided in their paper. In terms of calculating the intensity of absorption lines, a degeneracy factor of 7/6 for odd J'' and 5/6 for even J'' is included in the absorption coefficient [45]. The symmetry of the rotational states with respect to nuclei exchange is labelled *s* for symmetric states and *a* for antisymmetric states [36, 46].

2.3 Identifying the main electronic states

It was Mulliken [37, 38] who realised the visible absorption bands of iodine belong to a ${}^3\Pi_{0u} \leftarrow {}^1{\Sigma_g}^+$ transition, later labelled B - X, instead of a ${}^1\Sigma \leftarrow {}^1\Sigma$ transition as was previously thought. He also noted that the upper state fell into Hund's case (c), so it is $\Omega = 0$ and the symmetry terms which are of significance. Wei and Tellinghuisen in 1974 [47] consolidated the large number of studies on the molecular parameters of the B-X system. They state that overlapping investigations in a short time frame with a lack of comparisons of results lead to inconsistencies in results. Using these parameters in part, Tellinghuisen

[2] The rotational state changes the total symmetry by a factor of $(-1)^J$. For a ${}^1\Sigma_g^+$ molecule in the ground vibrational state, the vibrational and electronic wavefunctions do not change the total symmetry under nuclear exchange.

later calculated Franck-Condon Factors (FCFs) for the low vibrational levels $v = 0 - 5$ which covers room temperature absorption and factors for select fluorescence series (a *series* in this case being bands terminating at the same vibrational level).

vibronic, vibratinal are 1 or 0

Franck-Condon Factors give the relative intensities of the vibrational transitions. The rate of transition between states is given by the Einstein A coefficient[3]. This depends on wavenumber and the rotational states involved, but also the overlap integral between the two states and the electric dipole strength:

Franck-Condon does not include mu, and mue - equilibrium should not depend on R

$$A \propto |\langle v'' \mid \mu_e(R) \mid v' \rangle|^2. \tag{1}$$

The right hand side can be represented by the product of the average transition strength $|\mu_e|^2$ and the squared overlap integral between the potentials of the two states, $|\langle v'' \mid v' \rangle|^2$ which is the Franck-Condon Factor [47]. The sum of the FCFs for a discrete spectrum sums to 1, the factors do not give absolute intensities. Quantifying absolute intensities requires knowledge of the electronic transition strength $|\mu_e|^2$. $|\mu_e|^2$ was only found for the iodine visible spectrum to a high accuracy ($< 5\%$) in 2011 also by Tellinghuisen [45]. The motivation for the study was the whole visible-IR absorption spectrum, of interest to study of the I_2 concentration in the atmosphere, which includes the mostly continuous $A^3\Pi_{1u} \leftarrow X$ and completely continuous $C^1\Pi_{1u} \leftarrow X$ transitions[4].

electrontronic

Brown in 1931 reported an infrared absorption system [51]. This was later designated the $A^3\Pi_{1u} \leftarrow X$ system, with a continuum overlap with the $B - X$ spectrum studied in [52] and the vibrational numbering studied in [53, 54]. A 1996 paper by researchers at the University of Waterloo (Appadoo, Le Roy and Bernath) and the Laboratoire Aimé Cotton (including Gersernkorn and Luc) [55] summarises the many studies on the $A - X$ system, and provides a comprehensive set of molecular constants with which 9552 rovibronic lines were assigned by the authors [56] from the infrared portion of the iodine atlas [57].

$D^1\Sigma^+_{0u}$ is an ion-pair state which has been spectroscopically observed for more than a century. The $D \rightarrow X$ absorption bands occur in the vacuum ultraviolet and are known as the Cordes system [58], with a bound-free continuum spectrum known since 1913 [59]. The flourescence spectra of the $D - X$ system can be probed by first exciting

[3]The A coefficient gives a rate of spontaneous emission and the B coefficient gives the rate of absorption. They are related through the simple equation [33]

$$A = \frac{8\pi\nu^3}{c^3} B$$

where ν is the frequency of the radiation associated with the transition, so only one needs to be given.

[4]This C state was labelled B'' by Huber and Herzberg [48] but to match the equivalent labelling of this state in other halogens was changed to C. The state which was originally labelled C ($^3\Sigma^+_{1u}$ in refs. [48, 49]) is given the label C' in ref [50]. In the comprehensive PEC diagram from [32], shown in Figure 3 it is unclear which state corresponds to the original C state as according to Mulliken [40] cross-referenced with [49] it should be a $1u$ state corresponding to the ab dissociation limit.

to the B state and then up to the D state [60]. A portion of the spectrum from [58] is reprinted in Figure 1. Note the transmission wavenumbers of $39\,000\,\mathrm{cm}^{-1}$ compared to the wavenumber region of the atlas of Luc and Gerstenkorn which ends at $20\,000\,\mathrm{cm}^{-1}$ to cover the entirety of the $B - X$ spectrum.

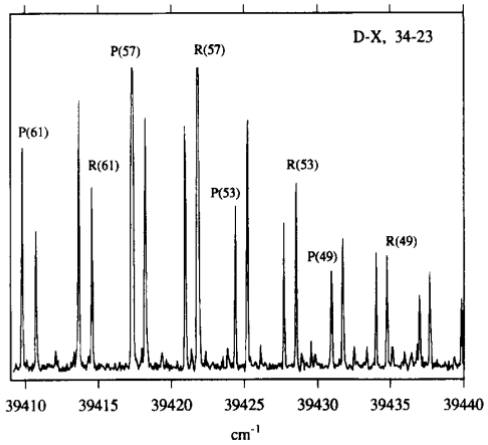


Figure 1: A portion of the (34, 23) band of the D-X absorption spectrum taken from [58]. Missing odd rotational states are a result of selection rules for collision energy transfer with an inert gas.

For an overall study of the bound states of I_2, Li and Balasubramanian [49] characterised 30 states of I_2 (including all 23 valence states) with electronic structure calculations. They provide potential energy curves for the lower lying states (see Figure 2), and both Hund case term symbols for each state, equilibrium internuclear separations, as well as the outer electron configurations following Mulliken's [40] labelling system. A later computation of just the valence states was conducted by Teichteil and Pelissier [61] which predict transition energies in closer agreement to experimental results than the Li and Balasubramian study.

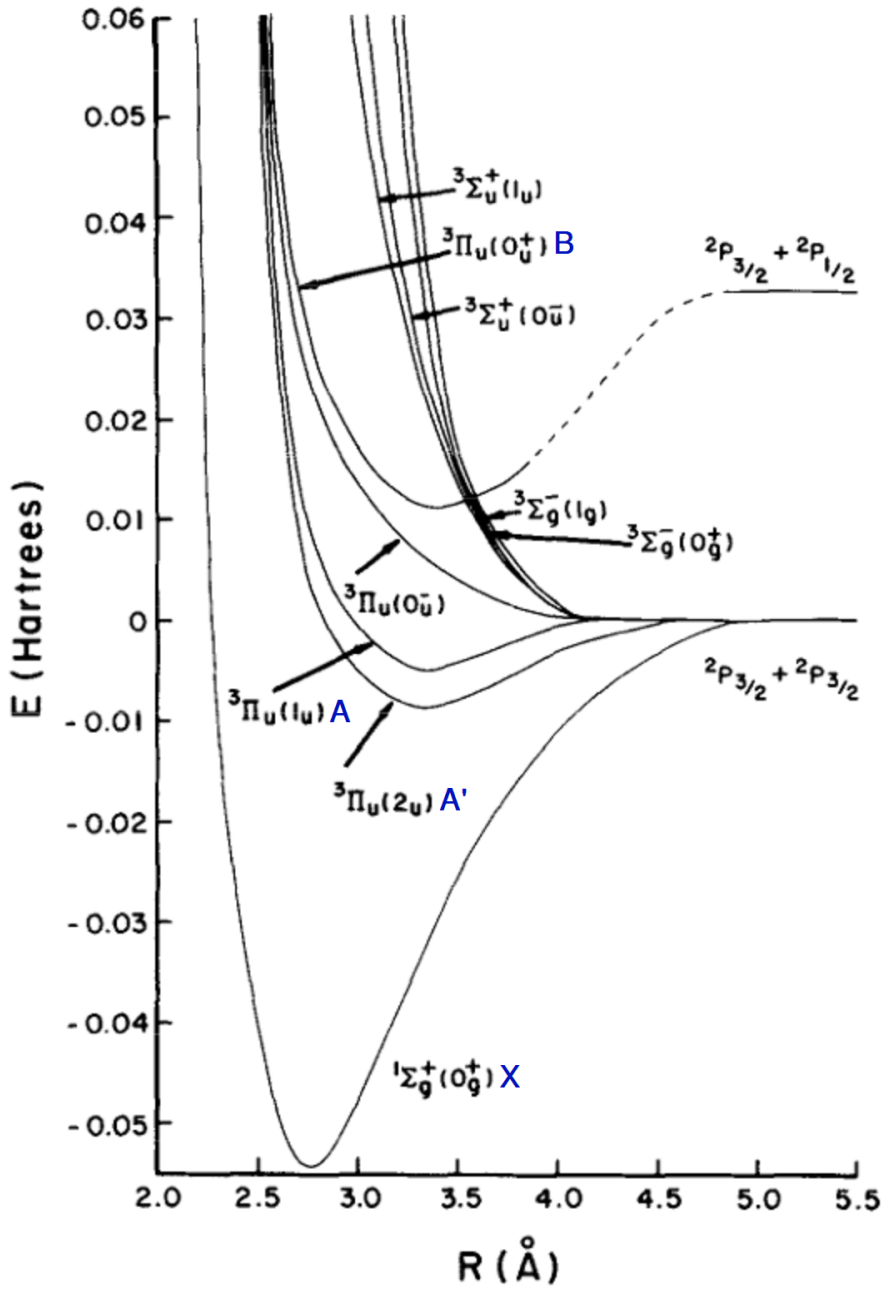


Figure 2: Potential energy curves for lower lying excited states of I_2 including spin-orbit effects, figure taken from [49]. I have added the Latin letter electronic state labels for the bonding states using assignments in the paper and checked for agreement with other literature cited in this paper. The only state shown corresponding to the second dissociation limit is the B state. Figure 3 shows more states but this diagram is included as it provides both sets of term symbols and the curves were directly calculated in the source paper.

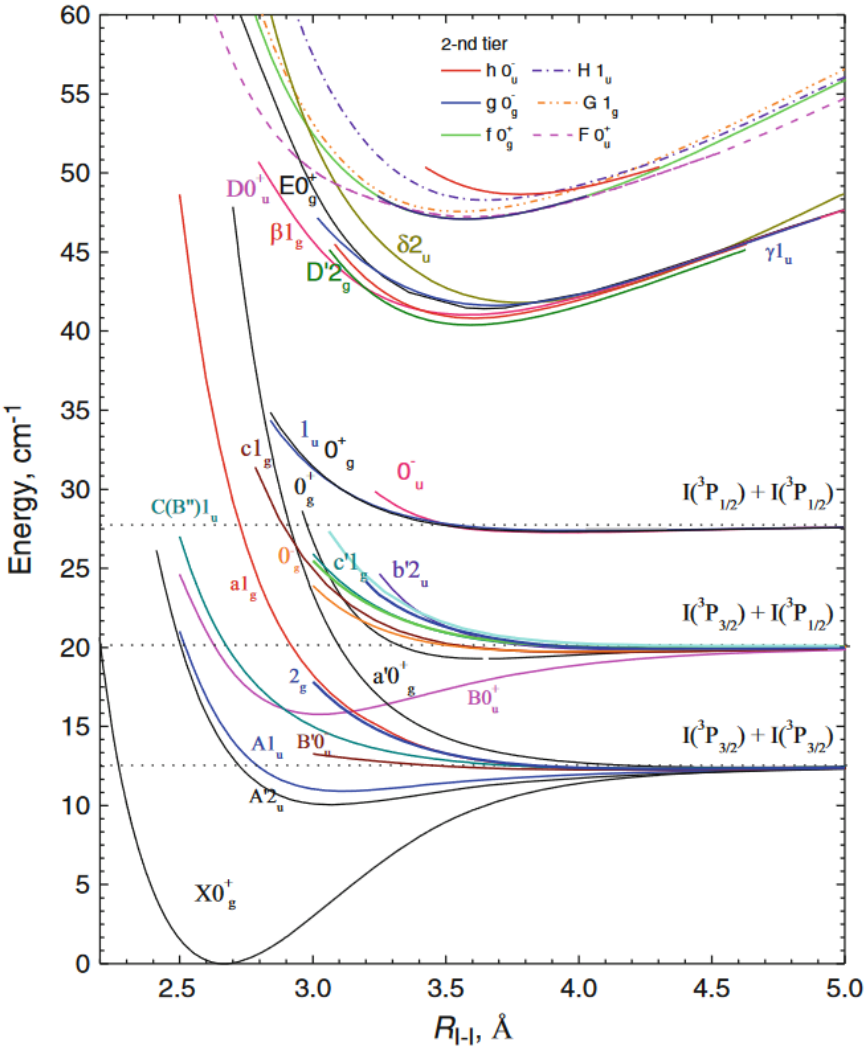


Figure 3: A diagram of potential energy curves taken from Pravilov *et al.* [32] showing 23 valence states with 3 sets of dissociation products, and some ion-pair states. States are labelled only in the Ω representation. Note that the vertical energy scale, from comparison with PECs in theoretical studies [49, 61], should be in $10^3\,\mathrm{cm}^{-1}$ not cm^{-1} as is given.

3 Spectroscopic Investigations of I_2

3.1 High-resolution spectral atlases

Gerstenkorn and Luc at the Laboratoire Aimé Cotton in the late 1970s published a series of iodine atlases using fourier-transform spectroscopy, with atlas II-III (14800-20000 cm^{-1}) covering nearly the full B-X spectrum [10]. A few regions of the spectrum in the vicinity of known laser lines were assigned soon after [62]. After a small systematic correction of $0.0056\,\mathrm{cm}^{-1}$, the wavenumbers in this volume have an estimated accuracy of $0.002\,\mathrm{cm}^{-1}$ [23]. This corresponds to a relative accuracy of one part in 10^7 [63]. A further, frequency dependent correction was published as part of a complement to the atlas in 1985 which included assignments of all the 14800-20000 cm^{-1} lines [11]. The lines were assigned by comparison with lines calculated using earlier derived spectroscopic parameters [5]. Lines in the 15780-15815 cm^{-1} region were measured later using a tunable laser in order to fill a 10 cm^{-1} gap caused by noise due to a monitoring He-Ne laser in the original study, as well as to compare tunable laser spectroscopy with the published Fourier-transform data [65]. Tellinguisen claims that as a result of the atlas of Luc and Gerstenkorn, the B-X system in iodine might be the most extensively studied diatomic transition with more than 10^5 assigned lines available at a precision of $10^{-3}\,\mathrm{cm}^{-1}$, sampling vibrational levels close to the respective dissociation limits [45].

It is the opinion of Pravilov *et al.* [32] that for calibration applications it is more convenient to use an iodine absorption spectrum in the form of the raw transition data, such as the spectrum of Salami and Ross [24], rather than a list of lines. To use this kind of data for the future purposes of a MARVEL project, it would be necessary to perform a peak finding routine on the data to identify transitions, and subsequently assign these lines. It is reported by the Salami and Ross that their spectrum can provide wavenumbers to an accuracy of 0.005 cm^{-1}, with the spectrum covering 14 250–20 000 cm^{-1}. Unresolved hyperfine structure is what determines the linewidths of 0.035-0.05 cm^{-1}. The iodine gas cell is brought up to 190°C to populate the states corresponding to the low wavenumber end. 200 lines were taken to calibrate the spectrum with the Gerstenkorn and Luc atlas as well as a similar transmission spectrum by Katô *et al.* [66, 67]. The atlas of Katô measures a similar wavelength region but resolves hyperfine structure. It is a transmission spectrum like that of Salam and Ross (ie. not a list of transmissions), but comes in the form of the raw spectrum alongside frequency markers. It is also not readily available online, and is to be ordered on CD-ROM format or a 4000 page paper version.

[5] Copies of all volumes of the atlas of Gerstenkorn and Luc were to be obtained from the Laboratoire Aimé Cotton, Bâtiment 505, Campus d'Orsay, 91405 Orsay Cedex, France [55] and are officially only available in paper format [24]. Scans of the main volume and the complement with assignments have been archived by the Laboratory of Physical Chemistry at the University of Athens [64].

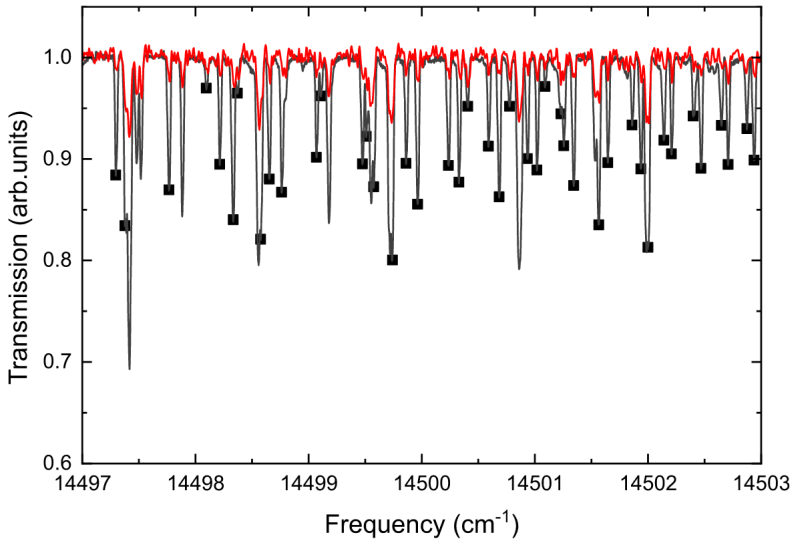


Figure 4: Figure from 23FeToCaKo [14] showing a 6 cm^{-1} wide portion of their spectrum in black. They give in red the spectrum of Salami and Ross [24] in red, noting a higher signal to noise ratio, and the black squares are peaks that are not found in Luc and Gersenkorn.

Recent high resolution studies of iodine from Torres *et al.* and Fernández *et al.* (with four authors in common between the papers), provided line lists in the 14600-14710 cm^{-1} [13] and 14400-14600 cm^{-1} [14] ranges. The authors used a tunable diode laser, measuring a combined 1852 absorption lines across both studies with a reported uncertainty of $3 \times 10^{-3}\,\mathrm{cm}^{-1}$. Results were compared with the works of Salami and Ross [24] and the Luc and Gerstenkorn (the 14000-15600 cm^{-1} atlas). They observed 238 new lines not reported by Luc and Gerstenkorn in the first work and 417 new lines in the second, some of which are shown in Figure 4. The data is readily available online but is a list of unassigned transitions. The wavenumber region does not overlap with the version of Luc and Gerstenkorn with the 1985 assignments but instead the 14000-15600 cm^{-1} atlas from 1980.

A similar laser absorption spectroscopic study is by Nöllecke *et al.* [12] covering the 10152-10929 cm^{-1} range (converted from nm) which is in the near infrared. They achieve a spectral resolution of 0.1 pm for Doppler-broadened lines of 1.5 pm width. The A-X system has a high laser power requirement, so the state saturation needed to achieve Doppler-free measurements could not be achieve with the laser used. The state populations were spread by heating the cell to 300°C, but the strongest lines had only 15% absorption. 9970 lines were identified, with an uncertainty given to be 0.15 pm or 50 MHz. Due to the Doppler broadening the underlying hyperfine structure is not resolved, the authors recommend further study with higher powered lasers in order to perform Doppler-free spectroscopy as this has not been performed for the A-X system.

They cite the previously mentioned Appadoo *et al.* [55] as the most recent analysis of the A-X system. The provided supplementary data is an unnasigned list of the transitions, the authors suggest in their work that rovibrational lines will be assigned to the primary data but a search of citations showed any follow up analysis that may have been conducted has not been published.

Table 2 lists spectroscopic investigations where as part of the published work or as supplementary material, a list of assigned and rotationally resolved transitions has been provided. Papers are named according to the year of publication and the first two letters of authors' surnames, up to the fourth author, with the full surname up to 8 characters for single author publications. The FTS studies of Gerstenkorn and Luc in the 70s still provide more lines over a broader frequency range than any other assigned iodine study, owing to the large spectral range available to Fourier Transform spectroscopy [68]. The laser spectroscopy studies previously discussed each cover a much smaller spectral range. Fernández *et al.* [14] recommend their laser absorption technique be used to study iodine at other spectral ranges, but this first requires finding a laser with a tuning range that encompasses the desired spectral range.

3.2 Hyperfine investigations

Hyperfine structure in rovibronic transitions is due to a change in the nuclear spin simulaneously with the change in electronic, vibration, and rotation levels. The is a change in F, the *total angular momentum inclusive of nuclear spin* [35]. The strength of the hyperfine interaction is determined by the magnetic dipole and magnetic quadrupole moments of the nucleus. Hanes and Dahlstrom in 1969 reported hyperfine splittings of the R(127) line of the (11, 5) B-X band at a coincidence with a He-Ne laser source (the 633 nm region mentioned in section 1.2). In the paper they suggested that the transition, which was measured to vary less than 2×10^{-13} wavelengths, could be used as a frequency reference. Hänsch *et al.* [88] two years later measured the hyperfine structure of a full rovibrational line using laser saturation spectroscopy. A review of calculated hyperfine constants was published in 1996 [89], the authors collected all the available fitted hyperfine constants. Constants had been found for 14 wavelengths (each corresponding to multiple tightly packed rovibronic lines), with each wavelength requiring four constants: the electric quadrupole eQq and constants for the spin-spin and spin-rotation interaction Hamiltonians.

An extensive hyperfine resolved study of iodine was a compilation of a decade's worth of measurements taken at the National Institute of Standards and Technology [63]. It provides hyperfine resolved measurements of 102 rovibronic lines, with typical uncertainty of 1 MHz. In another extensive hyperfine study for the purposes of frequency calibration

How many lines?

Table 2: Studies with measured and assigned spectroscopic data for I_2.

Source	System	Wavenumber range cm^{-1}
10Williams [69]	B-X	
97YuNiSu [70]	A-X	11200 - 12450
96ApLeBeGe [55–57]	A-X	
93UbAbMiSo [71]	B-X	18786 - 18788
93IsTaOb [72]	D-A	35246 - 41624
87VeChGeRa [73]	B-X	
87BrHo [74]	D-X	
86DiPaStKh [75]	B-X	
85DiPaKhAl [76]	B-X	
85GeLuAtlas [11]	B-X	14800 - 20000
82CoWe [77]	B-X, C'-B	15527 - 15620
80TeMcSu [78]	D-X	42671 - 50816
79TaDaGi [41]	D-X	17911 - 18602 (see appendix entry)
79BrCrHa [79]	B-X	
78ClMc [80]	B-X	
77SiHo [44]	B-X	17500 - 19500
77GeLuPe [81]	B-X	
77DaKi [82]	B-X	
75PaDwHo [83]	B-X	
74WeTe [47]	B-X	
73SiTe [84]	B-X	
69StCaWe [85]	B-X	
64BrKl [86]	B-X	
60Verma [87]	D-X	

[16], 1584 wavenumbers are calculated, but around 100 lines are measured experimentally. It is these lines that will be of use to a future empirical energy level study.

Lines determined from the Katô atlas mentioned in section 3.1, along with frequencies of specific hyperfine lines and the Gersenkorn and Luc atlas to fill in the gaps, 1500 rovibronic lines were collected and assigned by Knöckel *et al.* [90]. Their paper provides a detailed summary of hyperfine resolved investigations up to the publishing date. The data were collected and assigned in order to construct a physical model to describe the spectroscopic data and predict unobserved transitions. The model produced was for the visible specrtrum, and provides an accuracy ten times that of the Luc and Gerstenkorn atlas. The element of interest to a MARVEL project would be the authors' assigned line list from the spectral atlas of Katô. The paper states that the 1500 rovibronic lines are hosted online, but the link provided is to the publishers website. A search for this data as of the time of writing has not been successful.

Table 3 presents papers which have studied hyperfine resolved rovibronic transitions

and give absolute frequencies for the transitions (absolute as opposed to measuring the just the relative frequency differences of the hyperfine lines). Some of the lines provided are measured to extremely high precision, with most of the given papers measuring frequencies to between 11 and 13 significant figures. Knöckel *et al.* is omitted due to the unavailability of the data.

Table 3: Hyperfine resolved spectroscopic studies of iodine.

Hyperfine Papers	Rovibronic Lines
24ChXiQiMi [91]	13
19NeIgKuMa [92]	6
16KoAkHoIn [93]	3
08PhPe [94]	2
05Chen [95]	79
04EnHaRaWo [96]	12
04DuTr [97]	4
01HoNeZiUd [98]	16
00BoKlWiKn [99]	4
00XuVaHoYb [100]	numerous
00SiRo [101]	6
00CoLiLaBa [102]	3
99EdBaGiRo [103]	1
99GaBaBiBe [104]	1
99YeRoPiLo [17]	10
98BoBoKnNi [105]	3
98BoKlKnTi [106]	4
98VeVaHoUb [16]	$\sim$100
97Sansonet [63]	102
95JuEiSwYe [26]	1
94KrBoKnTi [15]	1
94ArBy [107]	1
93Razet [108]	1
93ArBy [109]	1
93AcZoAbRo [110]	1
93ShGiCoLi [111]	7
89MoHeSuIt [112]	2
71HaLeSc [88]	1
69HaDa [25]	1

4 Conclusion and Future Scope

This literature review has covered the some of the historical developments in understanding the electronic states in molecular iodine and spectroscopic studies of its main systems. The motivation is that iodine is a possible replacement for xenon as a propellant in electric space propulsion systems and spectroscopic information for I_2 is needed for modelling purposes; a wealth of spectroscopic information published in various forms is available for iodine for the completely unrelated purpose of frequency calibration. The future scope of this project is to connect the two ends of study. The MARVEL process put forward by Furtenbacher *et al.* [1] is an example of connecting a measured quantity, transition frequencies, with a quantity not directly measurable but of interest theoretically, in this case a network of rovibrational energy levels. Importantly it comes with a method of propagating uncertainties. The transitions are inverted to energy levels using a list of linear equations:

$$\mathbf{Ax} = \mathbf{y}. \tag{2}$$

Where **y** is a vector of all the transitions in energy units (dimension N_t where t is the number of unique measured transitions), **x** to be solved is a vector of dimension $N_1 - 1$ containing the energy levels. **A** is an $N_t \times (N_1 - 1)$ matrix with elements:

$$A_{ij} = \begin{cases} -1 & \text{if } E_j \text{ is the lower level of the } i\text{th transition} \\ 1 & \text{if } E_j \text{ is the upper level of the } i\text{th transition} \\ 0 & \text{otherwise.} \end{cases} \tag{3}$$

which connects the transitions and energy levels into a spectroscopic network.

The next steps of this project depend on the availability of the relevant data. If the available unassigned transitions outweigh the available assigned transitions to a sufficient degree either in accuracy or in quantity, then the next step would be to assign those transitions before the MARVEL process is conducted. Depending on the outcome of this, the future scope would be to use the rovribronic energy levels and existing potential energy curves in producing a line list using the software Duo [113]. The problem of unassigned transitions is a case of mismatched motivations for spectroscopic investigations. The MARVEL process and the use of Duo to predict transitions requires theoretical knowledge of measured transitions. Many of the iodine studies do not give assignments these are not necessary for the calibration tasks which motivate the investigations in the first place. All that is required for these tasks are line positions and relative intensities, such

as in refs. [12–14], or even spectra without explicitly determine peaks, refs. [24, 66]. The solution to this problem, as previously mentioned, would be to assign these spectra. There are also many studies, some included in the appendix, for which spectra were recorded and implicitly assigned in order to calculate spectroscopic constants or dipole transition strengths, but a list of transitions were not included either due to not being relevant to the main outcomes of the study or for space reasons. This problem turns the future task to that of archival research. Authors of sufficiently recent studies may have copies of the assigned line lists. Prof. Peter F. Bernath as already kindly provided the assigned list used in the Appadoo *et al.* study of the A-X spectrum [55, 56].

Appendix: Notes on relevant papers

In this appendix I provide brief notes on the spectroscopic studies which either provide an assigned or unassigned measured transition list; studies where reference is made to a transition list being measured but which is not provided; and studies where only vibrational bands without rotational lines is recorded. A similar summary of data to this section is commonly provided in MARVEL papers (see recently [114] for example), but instead of information on choices to be made later on in the MARVEL process such as uncertainties I will focus on the motivations and results of each study where relevant. Whilst ideally the list of papers in table 2 as will be used in a MARVEL study will be comprehensive, it is not claimed that this appendix contains every paper that *might* have a desired line list but ultimately does not. The focus is on studies which a) I have not yet discussed and b) which are often cited in the papers already discussed.

60Verma [87]: A study of the flourescence spectrum between the ground electronic state and an, at the time, uncharacterised upper state. Verma proposes that the transitions measured correspond to the same transition as the Cordes system which would place the upper state as the D $^3\Sigma_{0u}$ state. Errors are provided in milli-Angstrom, between ± 1 and $\pm 2\,\mathrm{m\AA}$.

64BrKl [86]: A fluorescence spectrum using sodium D lines to excite iodine to the B state. Assigned rovibronic lines are reported.

73WeTe [84]: According to the authors this was the first high-resolution work on the B-X emission spectrum due to weak B-X emission lines. The paper includes a useful table of known monochromatic source coincidences with iodine B-X lines.

75PaDwHo [83]: A fluorescence B-X study using excitation from an argon laser. The lines were assigned using the parameters of Wei and Tellinghuisen, but Tellinghuisen noticed a mistake in these assignments and published a corrected set of assignments [115].

76KaClMc [116]: Emission B-X study using circularly polarised argon lasers, the polarisation of the emission radiation was used to rotationally assign the spectrum. The lines themselves are not provided.

77GeLuPe [81]: A single B-X band was recorded and rotationally assigned with reference to a uranium standard.

78ClMc [80]: Rovibronic transitions around a 5145 $\mathrm{Angstrom}$ argon laser are recorded.

79Ashby [53]: Absorption measurements of the 9600 Å - 8000 Å region to study the A-X spectrum. Rovibrational lines and assignments are not given, only some vibrational band origins.

79BrCrHa [79]: Demonstration of a system of polarised lasers used to produce a sparse and readily assignable spectrum from only a few rotational states. Only 26 transitions were measured.

79TaDaGi [41]: A two photon technique is used to excite from the ground state to what they label as the T state, but which Brand and Hoy (see 87BrHo [74])reassign as the D0_u^+ state. The two photon technique results in an S branch due to ΔJ transitions being allowed. The wavenumbers given are from the laser frequency used to provide the two photon resonance, which is why the wavenumber range does not line up with other D-X spectra.

80AsJo [54]: The absorption spectrum was measured over a longer range than the previous study by one of the authors (79As) in order to provide more vibrational band numbering and locate the A state more reliably. The lines are not provided.

81ViSuTe [117]: A study of emission from a 1_g ion-pair state down to the A state. Only band

origins are resolved, with discussion of shading due to unresolved or uncharacterised rotational branches.

79GeLu, 81GeLu [62, 118]: Assignments of the B-X iodine atlas near a several common laser frequencies. The authors assigned the entire atlas a few years later, superseding these studies.

82David [119]: An A-X emission spectrum study, but only vibrational bands are measured.

82KoSiBa [120]: A study of the D'-A' spectrum. 50 A' vibrational states are rotationally analysed but the locations of the rotational lines are not provided.

82Tellingh [121]: A study of the D'-A' spectrum. Also no rotationally resolved lines provided.

82CoWe [77]: C'-X transitions via the B state are measured. Only a handful of transitions are measured, all originating from the same vibrational state.

83ViTe [122]: Emission from a tentitavely assigned ion-pair state. Only vibrational bands given.

85DiPaKhAl, 86DiPaStKh [75, 76]: Coherent anti-Stokes Raman spectroscopy is used to assign B-X transitions.

86MaBaChVe [123]: A flourescence spectra was recorded over the $7400\,\mathrm{cm}^{-1}$ to $20\,000\,\mathrm{cm}^{-1}$ range (converted from μm). Data was collected and analysed over the course of multiple publications [120, 124–127], characterising the long range behaviour or providing molecular parameters of the X, B, a, a', D' and A' states. Constants for the ground state up to the $v = 107$ state extend the study of [87] by an extra 23 vibrational states. However no rovibronic transition list is provided.

87BrHo [74]: reassignment of lines from Tai *et al.* [41] (see 79TaDaGi).

87VeChGeRa [73]: Photoacoustic spectroscopy of the 17580-18850 range of the B-X spectrum. The lines are assigned and compared to the Gerstenkorn and Luc atlas, which are of a higher precision.

92ZhFeHeTe [128]: A D'-A' absorption study. The 0.08 cm^{-1} laser resolution was resolved rotational structure which was analysed but only band origins are given.

93IsTaOb [72]: Two photon excitation study of the D state via the A and B states. Transitions are given and are in the vacuum-ultraviolet range.

93UbAbMiSo [71]: a single rovibrational level of the $\mathrm{E0}_g^+$ state. The actual frequency measurements of the flourescence to the D and B level are not given. What is given are the B-X transitions excited with an Nd:YAG laser in order to eventually excite up to the E state. The energy of the E state above the ground state is calculated not measured directly.

94LaBaCeCh [129]: Laser induced fluorescence study of the B-X system using FTS in order to study the variation of the transition dipole moment $|\mu_e|^2$ across different R-centroid values. A spectra was recorded at a $0.15\,\mathrm{cm}^{-1}$ and resolved rotational structure but the transition list is not provided.

97YuNiSu [70]: A-X spectrum is recorded between in the 11200-12450 cm^{-1} range. Q branch frequencies are given with reported 2.5σ uncertainties of around 0.005 cm^{-1}.

97CeBaChIn [130]: A study of the D'-A' spectrum. The clear reference to measured rotationally resolved transitions but again these are not explicitly given.

00XuVaHoYb [100]: Hyperfine lines in the B-X spectrum are measured for 15 vibrational bands with a 1σ uncertainty of 1.0 MHz.

02BoKnTi [131]: A hyperfine resolved study of an IR portion of the B-X spectrum. The recorded spectrum was used along with measured values in the literature (much of which makes

up Table 3 above) to provide hyperfine parameters for some rovibrational lines. The transitions recorded specifically for this study were not provided.

04ExBa [132]: A study of the D'-A' and D-A flourescence spectra. A spectrum was recorded but transitions are not provided in the paper. The study focused on the collisional transfer between the singlet D and triplet D' states in inert buffer gases.

05AkNoPoPr [133]: A study of emission transitions between the high-lying ion-pair states in order to calculate their transition dipole moment functions. The transition list is not given.

10Williams [69]: A study of B-X fluorescence induced by different He:Ne laser frequencies. The absorption frequencies are given, but the actual fluorescence measurements which are far more numerous are purported to be hosted by the publisher but appear to no be longer hosted on the publisher's website.

11Tea [45]: The spectrum 520–640 nm is recorded at 0.1 nm resolution to study the electronic transition moment $|\mu_e|^2$. A transition list is not provided.

13AkBaLuPo [134]: Excitation to an ion-pair state to calculate spectroscopic constants and PECs but raw spectroscopic data not provided.

14HrAcBuCh [135]: A study of the peak-to-peak widths of some hyperfine lines and relative frequencies between them, but absolute frequency measurements are not given.

18PeHu [136]: A study of the temperature dependence of the iodine spectrum. No transition list is provided.

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